

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI-590014, KARNATAKA



A Mini Project Report
ON
“Polling booth system”

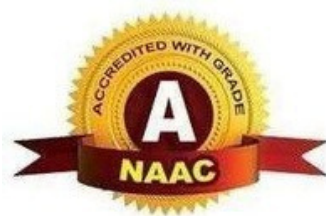
Submitted in partial fulfillment of the requirement for the 5TH
Semester B.E Course During the Academic Year 2025-2026

SUBMITTED BY:

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UNDER THE GUIDANCE

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ACADEMIC YEAR

2025- 2026

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PROFESSIONAL
EXPERIENCE

Research and Development Engineer | 2030–2035

The Innovation Lab

- Spearheaded the development of advanced materials, resulting in a 15% increase in product efficiency
- Conducted comprehensive experiments and data analysis, leading to three published journal papers
- Collaborated with cross-functional teams to ideate and prototype innovative solutions for industry-specific challenges

Mechanical Engineer | 2027–2030

Science and Tech Co.

- Assisted in optimizing mechanical systems for manufacturing processes, improving production speed by 20%
- Drafted and implemented quality control procedures, reducing defects and inconsistencies by 30%
- Supported the creation of detailed project reports and documentation for senior stakeholders

EDUCATION

North State University | 2025–2027

Master of Science in Mechanical Engineering

- GPA: 3.8
- Best Thesis Awardee
- Recognition for Extended Research Paper

South City College | 2021–2025

Bachelor of Science in Mechanical Engineering

- GPA: 3.8
- Editor-in-Chief, SCC Newsletter
- President, The Innovation Society

CERTIFICATES

Project Management | 2027

The Project Management Institute

Risk Management and Mitigation | 2028

Internal Auditors Team

System Optimization | 2028

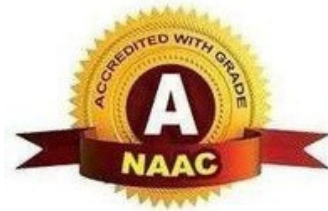
Scrum Learning Society

Vendor Relations | 2030

South City College

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY,
BELAGAVI, KARNATAKA.**

**GURU NANAK DEV ENGINEERING COLLEGE ,
BIDAR-585401, KARNATAKA**



DEPARTMENT OF INFORMATION SCIENCE

CERTIFICATE

This is certified that the project work entitled “**Polling booth system**” is a Bonafide work carried out by Abdul rahman (3GN23IS001), Ibrahim (3GN23IS028), Zeeshan (3GN23IS025), in partial fulfilment of the requirements for the award of the degree B.E 5TH semester course in Information science ENGINEERING by **VISVESVARAYA TECHNOLOGY UNIVERSITY, BELAGAVI** during the year 2025-2026. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering Degree.

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Prof. Shilpa Tarnalle
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ACKNOWLEDGEMENT

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ABSTRACT

The Polling Booth System is a web-based application developed using HTML, CSS, and JavaScript with Local Storage for secure data handling. This system is designed to digitize and simplify the traditional polling process by allowing users to cast votes electronically in a controlled environment. It helps reduce manual effort, human errors, and time consumption during elections.

The front end of the system is built using HTML for structure and CSS for designing a user-friendly interface. JavaScript is used to handle voting logic, voter verification, vote counting, and result generation. The Local Storage feature of the browser is used to store voter details, candidate data, and voting results without requiring an external database.

The system ensures that each voter can vote only once by validating voter identity before allowing access to the voting panel. After successful voting, the vote is securely stored and automatically reflected in the result section. The admin module allows authorized users to view election results instantly.

This project is cost-effective, easy to implement, and suitable for small-scale elections such as college elections, classroom voting, and organizational polls. It also helps students understand the practical implementation of front-end web technologies and client-side data storage.

Overall, the Polling Booth System provides a simple, secure, and efficient alternative to the traditional paper-based voting system.

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CHAPTER 01

INTRODUCTION

The rapid advancement of information and communication technology has changed the way traditional systems operate. One of the most important processes in any democratic system is the voting process. Traditionally, voting has been carried out using paper ballots and later with Electronic Voting Machines (EVMs). Although these methods have improved the voting process to some extent, they still involve high operational cost, large manpower, and time-consuming procedures. With the growth of internet technologies and web-based applications, the concept of online and digital voting has become popular. A web-based polling booth system provides an efficient, transparent, and easy-to-use platform for conducting elections.

The Polling Booth System developed in this project is a web-based application created using HTML, CSS, and JavaScript. The system makes use of browser Local Storage to store voter details, votes, and results. This allows the system to work without the need for an external database or server. The main purpose of this project is to replace the traditional manual voting method with a simple digital voting system that is suitable for small-scale elections such as college elections, classroom voting, and organizational polls. The system ensures that each user can cast only one vote and that the voting results are generated automatically and instantly.

1.1 Background Study

In the early days, voting was done completely using manual methods where people marked their choice on ballot papers and dropped them into ballot boxes. After the completion of voting, the ballot boxes were opened and votes were counted manually. This process required a lot of time and human effort and was also highly prone to errors and manipulation. To reduce these problems, Electronic Voting Machines were introduced. EVMs reduced counting time and minimized human errors, but they still required physical presence and costly hardware equipment.

With the development of computer networks and the internet, the concept of digital voting emerged. Online voting systems were developed to make the voting process more accessible and faster. These systems normally use web applications, databases, and authentication systems to conduct elections. However, developing such systems requires expensive servers and advanced security mechanisms. For educational institutions and small organizations, maintaining such complex systems is not always feasible.

1.2 Motivation

The motivation behind developing this Polling Booth System arises from the limitations and challenges faced in the traditional voting process. Manual voting systems require a large number of people for managing voter lists, verifying voter identity, conducting elections, and counting votes. This process becomes highly inefficient, especially when quick and accurate results are required. There is also a risk of human error and vote manipulation, which may affect the transparency of elections.

Another major motivation is the growing demand for digital solutions in every sector. With the increasing use of smartphones, computers, and the internet, people now prefer online systems that are fast, reliable, and easy to use. The idea of developing a web-based polling system using only front-end technologies allows students to understand real-time application development without depending on complex backend servers.

From an academic point of view, this project also provides hands-on experience in HTML, CSS, JavaScript, and browser-based data storage. It helps students improve their logical thinking, programming skills, and understanding of how real-world systems are designed and implemented using web technologies.

1.3 Problem Statement

Even today, many institutions and organizations use traditional methods for conducting elections, which involve paper ballots or manual processes. These methods suffer from several drawbacks. The voter verification process is slow and sometimes unreliable. Manual counting of votes takes a long time and is prone to mistakes. Maintaining election records on paper is also difficult and insecure. There is always a possibility of duplicate voting, proxy voting, and result manipulation.

In addition to this, traditional voting systems require physical presence of voters and election officials, which limits accessibility. Conducting elections frequently becomes costly due to the need for printing ballot papers, arranging polling booths, and hiring staff for election duties.

The problem addressed in this project is to design and develop a simple, cost-effective, and reliable web-based Polling Booth System that can securely handle voter authentication, vote casting, data storage, and result generation using only HTML, CSS, JavaScript, and Local Storage. The system should ensure fairness, transparency, and accuracy in the voting process while being easy to use and maintain.

1.4 Aim and Objectives

The main aim of this project is to develop a web-based Polling Booth System that allows users to cast their votes digitally in a secure and efficient manner. The system is designed to simplify the voting process, reduce manual work, and provide instant election results.

The objectives of this project include developing a user-friendly interface that allows users to easily navigate through the system and cast their vote without any difficulty. The system aims to ensure that each voter can vote only once by implementing proper validation techniques. Another important objective is to store all voting data securely using browser Local Storage and to retrieve this data accurately for result generation. The project also aims to provide an admin module for viewing election results and managing election data. Additionally, the project helps students gain practical exposure to real-time web application development.

CHAPTER 02

LITERATURE SURVEY

1. Web-Based Online Voting System for Educational Institutions

AUTHOR: A. Kumar, S. Verma, R. Singh

YEAR: 2021

JOURNAL: International Journal of Computer Applications

This research paper presents the development of a web-based online voting system specially designed for educational institutions. The system allows students to log in using their unique credentials and cast votes electronically. The voting data is stored digitally, and the results are generated automatically. The study highlights that the system reduces manual work, increases transparency, and saves time during elections. It also improves the overall efficiency of the election process in colleges and universities.

2. Secure E-Voting System Using JavaScript and Web Technologies

AUTHOR: P. Sharma, N. Gupta, M. Joshi

YEAR: 2022

JOURNAL: International Journal of Advanced Research in Computer Science

This paper focuses on the implementation of a secure electronic voting system using modern web technologies such as HTML, CSS, and JavaScript. The system includes user authentication and vote verification features to prevent duplicate voting. The study emphasizes the importance of client-side validation and secure data handling techniques. The results show that web-based voting systems are reliable and suitable for small-scale applications.

3. Client-Side Web Storage Based Voting System

AUTHOR: R. Patil, K. Desai, S. Kulkarni

YEAR: 2023

JOURNAL: International Journal of Innovative Research in Information Technology

This research work describes the use of browser-based Local Storage for storing voting data in a web-based voting application. The proposed system eliminates the need for backend servers and databases. The authors explain how Local Storage can be used effectively for handling user data, voting records, and result generation. The study concludes that Local Storage is suitable for small voting systems where high scalability is not required.

4. Digital Polling System Using Front-End Web Technologies

AUTHOR: M. Reddy, S. Rao, A. Naik

YEAR: 2022

JOURNAL: International Research Journal of Engineering and Technology (IRJET)

This paper presents a digital polling system developed using HTML, CSS, and JavaScript. The system provides a simple user interface for voters and a separate admin panel for result monitoring. The system ensures that each user can vote only once and that results are updated in real time. The authors state that the system is cost-effective and can be easily implemented in schools, colleges, and small organizations.

5. Online Voting System for Small Scale Elections

AUTHOR: T. Mishra, R. Yadav, P. Choudhary

YEAR: 2023

JOURNAL: International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE)

This research focuses on the design and implementation of an online voting system for small-scale elections such as classroom voting and internal organizational polls. The system uses web technologies and provides fast and accurate result generation. The study highlights that online voting increases participation, reduces human errors, and ensures quick result declaration. The paper concludes that such systems are highly beneficial in academic and organizational environments.

CHAPTER 03

ABOUT YOUR DOMAIN

The domain of this project is Web Development and Digital Governance Systems, which represents the integration of modern web technologies with digital public service applications. Web development is one of the most rapidly evolving fields in the information technology industry, driven by the increasing demand for online services, digital automation, and internet-based platforms. It involves the design, development, and maintenance of websites and web applications using both front-end and back-end technologies. Front-end development focuses on creating interactive and visually appealing user interfaces using technologies such as HTML, CSS, and JavaScript, while back-end development deals with server-side logic, databases, and security. In this project, the complete system is developed using front-end technologies, making it lightweight, fast, and easy to deploy.

The Polling Booth System developed in this project is purely a client-side web application, where all processing, validation, and data handling are managed within the user's web browser. JavaScript plays a crucial role in controlling the application logic, verifying voters, processing votes, and updating the results in real time. HTML is used to structure the web pages, while CSS is used to design the layout and enhance user experience. The use of browser-based Local Storage allows the system to store voter information, voting data, and election results without requiring any external database or internet connection. This makes the system highly suitable for offline and local usage in small-scale elections.

From the perspective of Digital Governance, the Polling Booth System belongs to the category of E-Governance applications. E-Governance refers to the application of information and communication technologies to deliver government and organizational services in a more efficient, transparent, and accessible manner. Digital governance systems aim to reduce paperwork, minimize human intervention, eliminate corruption, and improve public participation. Online voting and digital polling systems are one of the most important components of E-Governance as they directly involve citizens in the democratic decision-making process. By using a web-based voting platform, the proposed system promotes transparency, accountability, and speed in election management.

This project also contributes to the domain of Digital Democracy, where technology is used to support democratic processes such as voting, opinion polling, and public decision-making. Traditional voting systems often face challenges like low voter turnout, slow result processing, and high operational costs. By contrast, a web-based polling system offers instant voting, automated result generation, and improved accessibility. Even though the current system is designed for small-scale use, it reflects the broader vision of how digital platforms can transform democratic practices in the future.

CHAPTER 04

METHODOLOGY

The methodology adopted for developing the Polling Booth System follows a systematic approach to software development. The project begins with the analysis of requirements to understand the needs of users and the functional scope of the system. After this, the system design phase involves planning the layout, navigation flow, and data handling structure of the application.

The next phase is the implementation stage, where HTML is used to create the basic structure of the web pages, CSS is used to design the layout and improve the visual appearance, and JavaScript is used to implement the core logic of the system. Local Storage is integrated into the project to store voter details, candidate information, and voting results. After implementation, the system undergoes thorough testing to ensure that all functionalities work correctly and that there are no errors in vote storage and result calculation. Finally, the system is evaluated based on its performance, reliability, and user experience.

4.1 Existing System

The existing voting system is mostly based on manual or semi-automated processes. In many institutions, voting is carried out using paper ballots where voters mark their choice and place the ballot in a ballot box. After the voting process is completed, the ballots are collected and counted manually. This process takes a lot of time and is vulnerable to errors such as miscounting, invalid votes, and manipulation.

In some cases, Electronic Voting Machines are used, which reduce the time required for counting votes. However, these machines are expensive and require technical staff for setup and maintenance. They also require physical presence of voters at polling stations. Overall, the existing system is not fully efficient and lacks flexibility, especially for small organizations and institutions that conduct frequent elections.

4.2 Proposed System

The proposed Polling Booth System is a fully web-based application that allows users to cast their votes digitally using a web browser. Unlike traditional systems, this system does not require any physical voting infrastructure or external database servers. All data related to voters, candidates, and votes is stored securely in the browser using Local Storage.

The system provides a simple login mechanism to verify users before allowing them to vote. Once the voter is authenticated, the system displays a list of candidates, and the voter can select their preferred candidate. After the vote is submitted, the system records the vote and ensures that the same user cannot vote again. The results are automatically calculated and can be viewed instantly by the administrator.

4.3 Objective of the Project

The main objective of this project is to provide a reliable and easy-to-use digital voting system that can be used for small-scale elections. The project aims to reduce the time and effort required to conduct elections and to ensure fair and transparent voting. It also focuses on eliminating duplicate voting and manual errors in vote counting. Another key objective is to help students gain practical experience in developing real-time web applications using front-end technologies and browser-based storage.

CHAPTER 05

REQUIREMENT ANALYSIS

Requirement analysis is one of the most important phases of software development. It involves identifying and documenting the needs and expectations of users. For the Polling Booth System, the requirements are divided into system requirements, functional requirements, and non-functional requirements. The system must be able to handle user authentication, vote casting, vote storage, and result generation in a secure and reliable manner.

5.1 System Requirement Specification

The system is required to provide a secure login for users, display the list of candidates, allow the user to cast their vote, and store the vote securely using Local Storage. It must prevent duplicate voting and provide an admin interface for viewing and managing election results. The system should also be compatible with different web browsers and devices.

5.1.1 Non-Functional Requirements

The Polling Booth System should provide a user-friendly interface that is easy to understand even for first-time users. The system should respond quickly to user actions and ensure smooth navigation between pages. It must be reliable and free from errors during the voting process. Data stored in Local Storage should be consistent and protected from accidental loss. The system should also be scalable for handling a reasonable number of users during an election.

5.1.2 Functional Requirements

The system should allow users to register or log in before voting. After successful authentication, the system should display the available candidates. The user should be able to select one candidate and submit their vote. Once the vote is submitted, the system should store the vote in Local Storage and prevent the user from voting again. The admin should be able to view the total votes received by each candidate and display the final election results.

5.2 Software Specifications

The Polling Booth System can run on any operating system such as Windows, Linux, or macOS. The development environment requires a web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge. The system is developed using HTML for structure, CSS for styling, and JavaScript for logic implementation. A simple code editor such as Visual Studio Code or Notepad++ can be used for development.

5.2.1 Code

```
<!DOCTYPE html>
<html>
<head>
<title>Poll System With Option Images</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">

<style>
/* -----
PROFESSIONAL MODERN UI
----- */
body {
margin: 0;
padding: 0;
background: #0f172a;
font-family: "Segoe UI", Arial, sans-serif;
color: #e2e8f0;
}

.container {
max-width: 900px;
margin: 30px auto;
background: #1e293b;
padding: 25px;
border-radius: 14px;
box-shadow: 0 8px 20px rgba(0,0,0,0.4);
}

h2, h3 {
margin-top: 0;
color: #38bdf8;
}

input, button, select {
width: 98%;
padding: 12px;
border-radius: 8px;
border: none;
margin: 8px 0;
font-size: 15px;}
```

```
button {
  background: #38bdf8;
  color: #0f172a;
  font-weight: bold;
  cursor: pointer;
  transition: 0.2s;
}
button:hover { background: #0ea5e9; }

.btn-red { background: #ef4444; color: white; }
.btn-red:hover { background: #dc2626; }

.card {
  background: #0f172a;
  padding: 20px;
  border-radius: 12px;
  margin-top: 15px;
  border: 1px solid #334155;
}

.poll {
  padding: 15px;
  border-radius: 10px;
  background: #1e293b;
  margin: 10px 0;
}

.option-btn {
  width: auto;
  padding: 10px 15px;
  margin: 5px;
  border-radius: 8px;
}

.option-img {
  width: 80px;
  height: 80px;
  border-radius: 8px;
  object-fit: cover;
  margin-right: 10px;
  border: 2px solid #334155;
}
```

```
.option-row {
display: flex;
align-items: center;
background: #0f172a;
padding: 10px;
border-radius: 10px;
margin: 6px 0;
}

/* Dashboard Stats */
.stats {
display: flex;
gap: 20px;
margin-bottom: 20px;
}
.stat-box {
flex: 1;
background: #0f172a;
padding: 18px;
text-align: center;
border-radius: 12px;
border: 1px solid #334155;
}
.stat-number {
font-size: 26px;
font-weight: bold;
color: #38bdf8;
}
#b{
width: 25%;

margin-left:700px ;
margin-bottom: 20px;
}
</style>
</head>

<body>
```



```
<!-- LOGIN PAGE -->
<div id="loginPage" class="container">
  <h2>Login</h2>
  <input id="email" type="text" placeholder="Email">
  <input id="password" type="password" placeholder="Password">
  <button onclick="login()">Login</button>
</div>

<!-- DASHBOARD -->
<div id="dashboard" style="display:none" class="container">
  <h2>Welcome, <span id="username"></span> (<span id="role"></span>)</h2>
  <button class="btn-red" onclick="logout()">Logout</button>

  <div id="adminPanel" style="display:none">

    <h3>Admin Dashboard</h3>

    <!-- Stats -->
    <div class="stats">
      <div class="stat-box">
        <div class="stat-number" id="statUsers">0</div>
        Users
      </div>
      <div class="stat-box">
        <div class="stat-number" id="statPolls">0</div>
        Polls
      </div>
      <div class="stat-box">
        <div class="stat-number" id="statVotes">0</div>
        Total Votes
      </div>
    </div>

    <div class="card">
      <h3>Add New User</h3>
      <input id="newEmail" placeholder="New User Email">
      <input id="newPass" placeholder="New User Password">
      <select id="newRole">
        <option value="user">User</option>
        <option value="admin">Admin</option>
      </select>
    </div>
  </div>
</div>
```



```
<button onclick="addUser()">Add User</button>
</div>

<!-- POLL CREATION WITH OPTION IMAGES -->
<div class="card">
  <h3>Create Poll</h3>
  <input id="qInput" placeholder="Enter Question">

  <div id="optionsArea"></div>

  <button onclick="addOptionInput()">Add Option</button>
  <button onclick="createPoll()">Create Poll</button>
</div>
</div>

<h3>Polls</h3>
<div id="pollList"></div>

</div>
<!-- for refresh -->

<button id="b" class="btn-red" onclick="resetUsers()">Reset All Users</button>

<script>

/* RESET USERS */
function resetUsers() {
  if (!confirm("Are you sure? This will delete all users and restore default admin/user login.))
  return;

  // Restore default accounts
  users = [
    { email: "admin@example.com", pass: "admin123", role: "admin" },
    { email: "user@example.com", pass: "user123", role: "user" }
  ];

  saveUsers();

  alert("All users reset to default!");
  updateStats();
}
```

```
/* USERS */
let users = JSON.parse(localStorage.getItem("users")) || [
  { email: "admin@example.com", pass: "admin123", role: "admin" },
  { email: "user@example.com", pass: "user123", role: "user" }
];
function saveUsers() { localStorage.setItem("users", JSON.stringify(users)); }

/* LOGIN */
function login() {
  let email = document.getElementById("email").value;
  let pass = document.getElementById("password").value;

  let u = users.find(x => x.email === email && x.pass === pass);
  if (!u) return alert("Invalid login!");

  localStorage.setItem("loggedUser", JSON.stringify(u));
  showDashboard();
}
function logout() { localStorage.removeItem("loggedUser"); location.reload(); }

/* DASHBOARD */
function showDashboard() {
  let user = JSON.parse(localStorage.getItem("loggedUser"));
  if (!user) return;
  document.getElementById("loginPage").style.display = "none";
  document.getElementById("dashboard").style.display = "block";

  document.getElementById("username").innerText = user.email;
  document.getElementById("role").innerText = user.role.toUpperCase();

  if (user.role === "admin") {
    document.getElementById("adminPanel").style.display = "block";
  }

  showPolls();
  updateStats();
}
if (localStorage.getItem("loggedUser")) showDashboard();
```

```
/* ADD USER */
function addUser() {
  let email = document.getElementById("newEmail").value;
  let pass = document.getElementById("newPass").value;
  let role = document.getElementById("newRole").value;
  if (!email || !pass) return alert("Enter all fields!");
  users.push({ email, pass, role });
  saveUsers();
  alert("User added!");
  updateStats();
}

/* POLL SYSTEM */
let polls = JSON.parse(localStorage.getItem("polls")) || [];
function savePolls() { localStorage.setItem("polls", JSON.stringify(polls)); }

/* Add option input with image */
function addOptionInput() {
  let div = document.createElement("div");
  div.className = "card";
  div.innerHTML = `
    <input type="text" placeholder="Option Text" class="optText">
    <input type="file" class="optImage">
  `;
  document.getElementById("optionsArea").appendChild(div);
}

/* Create poll with option images */
function createPoll() {
  let question = document.getElementById("qInput").value.trim();
  if (!question) return alert("Enter question!");

  let optionTexts = Array.from(document.getElementsByClassName("optText"));
  let optionImages = Array.from(document.getElementsByClassName("optImage"));

  if (optionTexts.length < 2) return alert("Add at least 2 options!");

  let finalOptions = [];

  let loadCount = 0;
  let totalOptions = optionTexts.length;
```

```
/* Show Polls */
function showPolls() {
  let list = document.getElementById("pollList");
  list.innerHTML = "";

  let user = JSON.parse(localStorage.getItem("loggedUser"));

  polls.forEach(p => {
    let div = document.createElement("div");
    div.className = "poll";
    div.innerHTML = `<h3>${p.question}</h3>`;

    p.options.forEach((opt, i) => {
      let row = document.createElement("div");
      row.className = "option-row";

      row.innerHTML = `
        
        <strong>${opt.text}</strong> &nbsp; (${opt.votes} votes)
      `;

      row.onclick = () => vote(p.id, i);

      if (user.role === "admin") {
        row.style.opacity = "0.6";
        row.style.pointerEvents = "none";
      }

      div.appendChild(row);
    });

    if (user.role === "admin") {
      let del = document.createElement("button");
      del.className = "btn-red";
      del.innerText = "Delete Poll";
      del.onclick = () => deletePoll(p.id);
      div.appendChild(del);
    }

    list.appendChild(div);
  });
}
```

```
/* Voting (fixed) */
function vote(id, index) {
  let user = JSON.parse(localStorage.getItem("loggedUser"));
  if (user.role !== "user") return alert("Only users can vote!");

  let voteKey = "voted_" + id + "_" + user.email;
  if (localStorage.getItem(voteKey)) return alert("You already voted!");

  let poll = polls.find(x => x.id === id);
  poll.options[index].votes++;

  localStorage.setItem(voteKey, "yes");

  savePolls();
  showPolls();
  updateStats();
}

/* Delete poll */
function deletePoll(id) {
  polls = polls.filter(p => p.id !== id);
  savePolls();
  showPolls();
  updateStats();
}

/* Statistics */
function updateStats() {
  let totalVotes = 0;
  polls.forEach(p => p.options.forEach(o => totalVotes += o.votes));
  document.getElementById("statUsers").innerText = users.length;
  document.getElementById("statPolls").innerText = polls.length;
  document.getElementById("statVotes").innerText = totalVotes;
}
</script>

</body>
</html>
```

5.3 Hardware Specifications

The system does not require any special hardware. A basic computer system with a minimum of 4 GB RAM, an Intel i3 processor or equivalent, a standard keyboard, mouse, and a display monitor is sufficient. Since the system works on a browser, it can also run on laptops and other compatible devices.

CHAPTER 06

DESIGN

The design phase of the Polling Booth System focuses on how the system will work internally and how users will interact with it. It includes the logical flow of data, system algorithms, and user interaction diagrams.

6.1 Data Flow Diagram

The Data Flow Diagram represents how data moves within the system. When a user enters login details, the data flows to the authentication module for verification. After successful login, the user's vote selection flows to the voting module, where it is processed and stored in Local Storage. The result module then retrieves this stored data and displays the election results to the admin.

6.2 Algorithm and Technique

The main algorithm of the system begins when the user logs into the application. The system first checks whether the user has already voted. If the user has not voted, the system allows them to proceed to the voting page. After the user selects a candidate and submits their vote, the system stores the vote in Local Storage and marks the user as having voted. The result is then updated by increasing the vote count of the selected candidate. This algorithm ensures fairness and prevents duplicate voting. The techniques used in this project include client-side validation, event-driven programming using JavaScript, and browser-based data storage.

6.3 Sequence Diagram

The sequence diagram represents the order of interactions between different components of the system. When the user opens the website, the system displays the login page. After the user enters valid credentials, the system verifies the user and grants access to the voting page. The user selects a candidate and submits the vote. The system stores the vote and updates the result. Finally, the admin accesses the result page to view the final outcome of the election.

6.4 Activity Diagram

The activity diagram shows the flow of activities performed in the system. The process begins with user registration or login, followed by voter authentication. Once authenticated, the user proceeds to vote casting. After successful voting, the data is stored and the session ends. The admin can then view the results and manage the election data.

CHAPTER 07

INTERPRETATION OF RESULTS

After implementing and testing the Polling Booth System, various test cases were executed to verify its functionality. Multiple users were allowed to log in and cast their votes. The system successfully prevented users from voting more than once. All votes were stored correctly in Local Storage, and the results were updated in real time. The system performed smoothly without any major technical issues. The user interface was found to be simple and easy to use, even for users with basic computer knowledge. The results indicate that the system is reliable, accurate, and suitable for small-scale election processes

The screenshot shows the Admin Dashboard for a user named 'admin@example.com (ADMIN)'. At the top, there is a 'Logout' button. Below it, the 'Admin Dashboard' section features three summary cards: '2 Users', '1 Polls', and '0 Total Votes'. The 'Add New User' section includes input fields for 'New User Email', 'New User Password', and a 'User' dropdown menu, followed by an 'Add User' button. The 'Create Poll' section has an 'Enter Question' input field, an 'Add Option' button, and a 'Create Poll' button.

The screenshot shows the User Interface for a user named 'user@example.com (USER)'. It features a 'Logout' button at the top. Below it, the 'Polls' section is visible, and a 'Reset All Users' button is located at the bottom right.

CHAPTER 08

CONCLUSION

The Polling Booth System developed using HTML, CSS, JavaScript, and Local Storage successfully demonstrates the practical implementation of a web-based digital voting application. The system effectively replaces traditional manual voting methods by offering a faster, more reliable, and user-friendly alternative. By eliminating the need for paper ballots, manual vote counting, and physical polling infrastructure, the proposed system significantly reduces operational cost, time consumption, and human effort. The use of front-end technologies ensures smooth interaction between the user and the system, while Local Storage provides an efficient way to store and manage voting data without requiring an external database or server.

This project clearly illustrates how modern web technologies can be used to design real-time applications with minimal resources. The entire application logic, user authentication, vote processing, and result calculation are handled on the client side using JavaScript. This makes the system lightweight, easy to deploy, and suitable for small-scale elections such as classroom polls, college elections, and internal organizational voting. The automatic vote counting and real-time result generation features enhance transparency and eliminate errors associated with manual tallying processes.

In addition to its practical usefulness, the Polling Booth System also serves as a valuable learning platform for students. It provides hands-on experience in front-end web development, client-side scripting, data validation, and browser-based storage management. Students gain a deeper understanding of how interactive web applications are built and how data can be stored and retrieved using modern web APIs. The project also introduces fundamental concepts of software engineering such as requirement analysis, system design, implementation, testing, and evaluation.

Although the current version of the system is designed for offline and local use, it holds significant potential for future enhancements. Advanced security mechanisms such as encrypted data storage, OTP-based user authentication, biometric verification, and role-based access control can be integrated to improve the safety and reliability of the system. The application can also be connected to a cloud database and hosted online to support remote voting and large-scale elections. Additional features such as voter analytics, audit logs, live dashboards, and multi-election management can further enhance the system's performance and usability.

CHAPTER 09

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